

Build a Paper Prototype

Project: Helpful Classroom Sorter

1. Recall and choose your starting point

Try the recall before opening notes. Then use a different colour or a labelled second line to correct your explanation.

Design the Decision Process

Without opening the old lesson, explain one key idea from Design the Decision Process. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

Find a Real Problem

Without opening the old lesson, explain one key idea from Find a Real Problem. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

Prompts Are Design Instructions

Without opening the old lesson, explain one key idea from Prompts Are Design Instructions. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

2. Practise with fading help

Use W2: Run and record the first match for Building and tracing a paper prototype. Record the answer once; the lesson and workbook refer to the same attempt.

Given inputs · Procedure I chose · Middle steps · Result and check

Given inputs: _____

Procedure I chose: _____

Middle steps: _____

Result and check: _____

The step I completed without help, and the hint I needed

3. Individual reasoning check

Use the worked case for Building and tracing a paper prototype. Proposed explanation: “A correct-looking answer is enough”. Decide whether this explanation is supported. Point to the step or supplied fact that decides; explain your repair if needed.

My explanation before partner discussion

4. Independent transfer case

Independent case: Build a new route through a small catalogue

Independently make and trace instruction cards for this paper equipment catalogue: E1 first, capacity 3 people; E2 second, capacity 6. Valid input is a whole group size from 1 to 8. Suggest the first item whose capacity is at least the group size; no booking occurs. Include invalid input and no match. Write expected results, then operate your cards for groups of 5 and 7. Record both runs and explain every stopping point.

My independent explanation, trace and limits

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

5. Your learning target

Assign three people as input reader, rule follower and output recorder. Run the repaired sorter on paper, metal and an unreadable description. Record expected and observed outcomes. What does this prototype demonstrate?

6. Solve the given case

A paper prototype has input cards, decision cards and paper, plastic and review trays.

How can a learner simulate it?

Show your trace, calculation or evidence

Write or draw your reasoning and evidence.

7. Check a changed case

Your paper prototype shows an input card, a decision card and an output card. Explain what a person must simulate and what the prototype cannot prove.

My independent answer and reason

Write or draw your reasoning and evidence.

Week 32 Test and reflect

8. Design a revealing test

Create a changed input, boundary case or missing-information case. Write the expected answer or safe response before testing.

Test record · My evidence

Test record: Changed input or condition

My evidence: _____

Test record: Expected answer or fallback

My evidence: _____

Test record: Observed result

My evidence: _____

Test record: Why the result occurred

My evidence: _____

9. Project checkpoint

Build the complete paper sorter: input cards, process guide, output cards, not-sure card, and test log.

The project change and evidence I will keep

Write or draw your reasoning and evidence.

10. Self check

I can explain the mechanism. I can show evidence. I can handle a changed case. I can name a limit. Mark each as not yet, with help or independently, and attach supporting work.

Teacher review _____ Next step _____

11. Transfer practice from the original programme

Assign three people as input reader, rule follower and output recorder. Run the repaired sorter on paper, metal and an unreadable description. Record expected and observed outcomes. What does this prototype demonstrate?

My independently reasoned answer

Record support used: none, hint or teacher explanation.

12. W1 Assemble the runnable kit

Create P1 version 1 using the three instruction cards in the lesson. Copy catalogue B version 1 and make a request card, current-row marker and trace sheet. List what each item records. Explain which information stays fixed during one run and what changes.

My reasoning and evidence

13. W2 Run and record the first match

Use the manual checklist to predict P1's output for 11 minutes No. Then ask a partner to operate your saved cards. Record the expected result separately from the observed steps and output, including the reason. Explain why the operator should stop without checking later rows.

My reasoning and evidence

14. W3 Locate the source of a wrong result

Saved instructions P0 say "start at B1; if B1 fails, report no match." For the valid request 9 minutes No, an operator follows P0 exactly and reports no match. Identify the instruction defect and write the changed step. Explain how to label the correction and what to retain from the failed run.

My reasoning and evidence

15. W4 Build a new route through a small catalogue

Independently make and trace instruction cards for this paper equipment catalogue: E1 first, capacity 3 people; E2 second, capacity 6. Valid input is a whole group size from 1 to 8. Suggest the first item whose capacity is at least the group size; no booking occurs. Include invalid input and no match. Write expected results, then operate your cards for groups of 5 and 7. Record both runs and explain every stopping point.

My reasoning and evidence

16. W5 Make the evidence repeatable

A teammate keeps only a final “B2” answer and throws away the request and old instructions after editing them. Explain why another person cannot check this result. List what to save for each run and explain why an observed error must remain in the record.

My reasoning and evidence

17. Feedback, return check and learning record

Evidence · My record

Evidence: Worked alone / hint / model

My record: _____

Evidence: First step I repaired and why

My record: _____

Evidence: New case checked later; date and result

My record: _____

Evidence: What I can now explain without notes

My record: _____

Evidence: One remaining question

My record: _____

Revisit this idea with your teacher after a delay. Familiarity is different from explaining and using it on a changed case.

18. Pictorial case practice: Build a Paper Prototype

A fictional team turns a storage-bin specification into movable instruction cards.

Bins X capacity 3 then Y capacity 7; valid count 1–9. Select first capacity at least count.

For request 4, write baseline Y before operating cards. Prototype cards mistakenly stop with X immediately.

Build a Paper Prototype

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Expected result

Baseline checks X then Y and expects Y for count 4.



2. Operate literally

Follow the current prototype cards without secretly repairing them.



3. Actual trace

Prototype stops at X; it never performs the capacity comparison.



4. Record mismatch

Expected Y, actual X: locate the missing check.

CHANGE ONE THING

Insert "if capacity is below count, continue" before accepting a bin.

Explain the changed result using the matching lesson.

What should appear in the test log before and after the repair?

My picture-based explanation and changed-case prediction

Support used: alone / hint / worked example. This record is practice evidence.

Project connection: Save sorter instruction cards, dated versions and operator traces alongside the manual expected outputs.

19. Project portfolio milestone

Project: Helpful Classroom Sorter

Current milestone: Present and defend. Demonstrate the system, explain one failure and justify the next improvement with evidence.

Save sorter instruction cards, dated versions and operator traces alongside the manual expected outputs.

Evidence to keep · My record

Evidence to keep: This week's input and deciding step

My record: _____

Evidence to keep: Prediction and actual result (or labelled paper trace)

My record: _____

Evidence to keep: One change since the previous checkpoint and why

My record: _____

Evidence to keep: One error or limit and the next test

My record: _____

Explain what this week adds to your project. Reuse last week's record rather than starting again.

20. Supported practice and learning evidence

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

A fictional team turns a storage-bin specification into movable instruction cards.

Practice 1: explain the deciding evidence

Which labelled step matches this detail? Expected Y, actual X: locate the missing check.

1. Actual trace
2. Record mismatch
3. Operate literally
4. Expected result

My selected conclusion, deciding fact and explanation

Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

After your attempt, compare with the supported explanation: Record mismatch Expected Y, actual X: locate the missing check. A paper prototype is executable when another person can follow its instructions without guessing your intention. Record the actual wrong result separately from the expected one. Discovering a missing comparison is useful evidence for repair; pretending the operator followed the intended rule would hide the defect.

Practice 2: explain the deciding evidence

Changed-case practice: Insert “if capacity is below count, continue” before accepting a bin. Which conclusion is supported by the supplied facts?

1. This one example guarantees correct results for every future input.
2. The revised trace rejects X and accepts Y for 4.
3. We can decide the result without examining the changed input or the procedure.

My selected conclusion, deciding fact and explanation

Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

After your attempt, compare with the supported explanation: The revised trace rejects X and accepts Y for 4. $3 < 4$ is true; $7 < 4$ is false, so Y fits.

Repair a missing building block

Review Design the Decision Process (ai-1/31). Goal: Design a decision process that covers valid, invalid and no-match requests. Short practice: Where does the first original rejection occur, and why is K3 not originally selected?. Return to this week and explain what you changed.

Review Find a Real Problem (ai-1/29). Goal: Describe a specific user difficulty using supplied evidence without inventing its cause. Short practice: Write a one-sentence project goal and say what evidence would count as a failure.. Return to this week and explain what you changed.

Review Prompts Are Design Instructions (ai-1/26). Goal: Write a prompt with a clear task, supplied evidence and checkable output requirements. Short practice: What must happen if B instead conflicts with another equally authoritative duration record?. Return to this week and explain what you changed.

Keep your completed work

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run evidence. Open the HTML copy to read or print it. Drafts stay in this browser when storage is available; clear drafts on a shared device.

What I repaired, the evidence I kept and the next test
